THE GYSIN MAPS ATTACHED TO TWO CARTIER DIVISORS COMMUTE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The Gysin homomorphism attached to a triple (invertible sheaf, section, zero scheme). Same chapter chow.tex, label definition-gysin-homomorphism, invoked in the statement at proof line 5500. Fixes what the four arrows in the square are, so that the commuting assertion has a meaning; the comparison itself is carried out in the counted span.
- The discussion of the key formula: local generators at generic points of divisors. Same chapter chow.tex, label section-key, invoked at proof lines 5534 and 5537. Supplies the normalisation of the meromorphic sections sigma and sigma' at each generic point and the definitions of the multiplicities n_i and n'_i, which is the setup of the computation.
- The key formula and infinite sums of rational equivalences. Same chapter chow.tex, labels lemma-key-formula (proof line 5556) and remark-infinite-sums-rational-equivalences (proof line 5568). Gives the cycle identity turning the difference of the two representatives into a sum of divisors of tame symbols, and lets the resulting rational equivalence pass from the classes [W] to an arbitrary cycle.

Source reference. Original lemma 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Navigation pointer for \ref{definition-gysin-homomorphism} used inside the counted span (line 5500); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYC: sources/stacks-chow/pinned-original-source.tex, source lines 4746–4746. Navigation pointer for \ref{lemma-key-formula} used inside the counted span (line 5556); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02RZ: sources/stacks-chow/pinned-original-source.tex, source lines 3057–3057. Navigation pointer for \ref{remark-infinite-sums-rational-equivalences} used inside the counted span (line 5568); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AYB: sources/stacks-chow/pinned-original-source.tex, source lines 4686–4686. Navigation pointer for \ref{section-key} used inside the counted span (line 5534); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5497); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ and $(\mathcal{L}', s', i' : D' \to X)$ be two triples as in Definition 02T8. Then the diagram*

$$\begin{array}{ccc} \mathrm{CH}_k(X) & \xrightarrow{i^*} & \mathrm{CH}_{k-1}(D) \\ {\scriptstyle (i')^*}\downarrow & & \downarrow{\scriptstyle j^*} \\ \mathrm{CH}_{k-1}(D') & \xrightarrow{(j')^*} & \mathrm{CH}_{k-2}(D \cap D') \end{array}$$

commutes where each of the maps is a Gysin map.

Proof. Denote $j : D \cap D' \to D$ and $j' : D \cap D' \to D'$ the closed immersions corresponding to $(\mathcal{L}|_{D'}, s|_{D'}$ and $(\mathcal{L}'_D, s|_D)$. We have to show that $(j')^* i^* \alpha = j^*(i')^*\alpha$ for all $\alpha \in \mathrm{CH}_k(X)$. Let $W \subset X$ be an integral closed subscheme of dimension k. Let us prove the equality in case $\alpha = [W]$. We will deduce it from the key formula.

We let σ be a nonzero meromorphic section of $\mathcal{L}|_W$ which we require to be equal to $s|_W$ if $W \not\subset D$. We let σ' be a nonzero meromorphic section of $\mathcal{L}'|_W$ which we require to be equal to $s'|_W$ if $W \not\subset D'$. Write

$$\mathrm{div}_{\mathcal{L}|_W}(\sigma) = \sum \mathrm{ord}_{Z_i, \mathcal{L}|_W}(\sigma)[Z_i] = \sum n_i [Z_i]$$

and similarly

$$\mathrm{div}_{\mathcal{L}'|_W}(\sigma') = \sum \mathrm{ord}_{Z_i, \mathcal{L}'|_W}(\sigma')[Z_i] = \sum n'_i [Z_i]$$

as in the discussion in Section 0AYB. Then we see that $Z_i \subset D$ if $n_i \neq 0$ and $Z'_i \subset D'$ if $n'_i \neq 0$. For each i, let $\xi_i \in Z_i$ be the generic point. As in Section 0AYB we choose for each i an element $\sigma_i \in \mathcal{L}_{\xi_i}$, resp. $\sigma'_i \in \mathcal{L}'_{\xi_i}$ which generates over $B_i = \mathcal{O}_{W,\xi_i}$ and which is equal to the image of s, resp. s' if $Z_i \not\subset D$, resp. $Z_i \not\subset D'$. Write $\sigma = f_i \sigma_i$ and $\sigma' = f'_i \sigma'_i$ so that $n_i = \mathrm{ord}_{B_i}(f_i)$ and $n'_i = \mathrm{ord}_{B_i}(f'_i)$. From our definitions it follows that

$$(j')^* i^*[W] = \sum \mathrm{ord}_{B_i}(f_i) \mathrm{div}_{\mathcal{L}'|_{Z_i}}(\sigma'_i|_{Z_i})$$

as cycles and

$$j^*(i')^*[W] = \sum \mathrm{ord}_{B_i}(f'_i) \mathrm{div}_{\mathcal{L}|_{Z_i}}(\sigma_i|_{Z_i})$$

The key formula (Lemma 0AYC) now gives the equality

$$\sum \Big(\mathrm{ord}_{B_i}(f_i) \mathrm{div}_{\mathcal{L}'|_{Z_i}}(\sigma'_i|_{Z_i}) - \mathrm{ord}_{B_i}(f'_i) \mathrm{div}_{\mathcal{L}|_{Z_i}}(\sigma_i|_{Z_i}) \Big) = \sum \mathrm{div}_{Z_i}(\partial_{B_i}(f_i, f'_i))$$

of cycles. Note that $\mathrm{div}_{Z_i}(\partial_{B_i}(f_i, f'_i)) = 0$ if $Z_i \not\subset D \cap D'$ because in this case either $f_i = 1$ or $f'_i = 1$. Thus we get a rational equivalence between our specific cycles representing $(j')^* i^*[W]$ and $j^*(i')^*[W]$ on $D \cap D' \cap W$. By Remark 02RZ the result follows for general α. $\square$